The University of Texas at Austin
Cockrell School of Engineering

ECE 374N

tACS + BCI EFFECTS ON WORKING MEMORY

Project I: EEG BCI + Transcranial Alternating Current Stimulation (tACS)

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HYPOTHESIS

Active tACS at individualized theta frequency over a medial frontal target will enhance Day 2 BCI learning compared with sham.

Marker: Frontal Midline Theta (FMT) power at electrode Fz, 4-8 Hz

Prediction 1: BCI Learning

Active tACS group shows greater FMT modulation and steeper improvement across BCI runs 1 → 2 → 3 → 4 vs. Sham.

Prediction 2: Entropy Change

Active tACS group shows greater change in entropy in N-back inter-trial baselines (1500 ms fixation, post vs. pre tACS) vs. Sham.

tACS → Theta Power → BCI Learning

Zeng et al., 2022; Yeh et al., 2022

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Rationale for tACS -- Why implement it?

- tACS delivers a weak oscillating current at each participant's individualized theta frequency over the dACC
- This is expected to entrain the brain's natural theta rhythm, especially priming the circuit before BCI neurofeedback on Day 2
- The sham conditions ramp up for only about 10 seconds then turns off, so participants cannot tell the difference → allowing us to isolate the true stimulation effect

tACS → Theta Power → BCI Learning

Zeng et al., 2022; Yeh et al., 2022

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Why Frontal Midline Theta (FMT)?

- **Definition: EEG oscillations at 4-8 Hz, maximal at Fz/FCz electrodes.**
- **Associated with medial frontal / dorsal anterior cingulate cortex (dACC)-related control processes.**

Tracks WM Load

- FMT power increases as N-back difficulty rises → theta synchronizes as cognitive demand grows
- Reflects sustained attention & cognitive control
- Most produced in N-back vs. WM tasks

(Hsieh & Ranganath, 2014)

Matches our Stimulation Target

- FMT is source-localized to the medial frontal cortex (MFC) and anterior cingulate cortex (ACC)
- tACS targets dACC at individualized theta frequency
- tACS is expected to directly entrain endogenous FMT

(Cavanagh & Frank, 2014; Botvinick et al., 2004)

Real-Time BCI Regulation

- FMT is a sustained oscillatory signal, computable trial-by-trial in real time
- Neurofeedback can increase FMT across sessions
- If tACS primes theta activity, BCI regulation of FMT should be enhanced (as mentioned in our prediction

(Yeh et al., 2022; Wang & Hsieh, 2013)

FMT is a strong candidate marker because it links WM demand, medial frontal control (FMT), and real-time EEG feedback.

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Why include Entropy?

Entropy = a measure of EEG signal unpredictability / complexity

Low		High
regular, repeating signal		complex, irregular signal

In EEG, entropy captures broadband complexity changes that traditional power spectral analysis (like theta power) might miss.

If tACS changes the brain's dynamics, entropy should detect it — even if theta power alone does not ().

Shannon Entropy

- $H = -\Sigma\, p(x) \log p(x)$

*p(x) = relative frequency (probability) of the EEG signal falling into amplitude bin x during a 1,500 ms baseline epoch.

-Measures **unpredictability of amplitude distribution**

Sample Entropy

- Measures regularity of EEG time series
  - **does the pattern repeat over time?**

Theoretical basis: Dynamic Network Biomarker (DNB) theory → systems approaching state transitions show increased entrop

(Tian et al., 2017; Bruce et al., 2009; Chen et al., 2012 ; Kang et al., 2018)

Why Entropy, not just FMT?

	FMT Power	VS	Entropy	
Signal Magnitude	Measures energy in one specific frequency band (4-8 Hz). Capture Δ theta amplitude	↔	Measures overall signal complexity across all frequencies.	**Signal Dynamics**
Limitation	Misses broadband complexity shifts, changes in signal regularity across frequencies.	↔	Capture nonlinear dynamics, regularity changes, state transitions.	**Strength**

(Hsieh & Ranganath, 2014; Tian et al., 2017; Lau et al., 2022)

Data Structure

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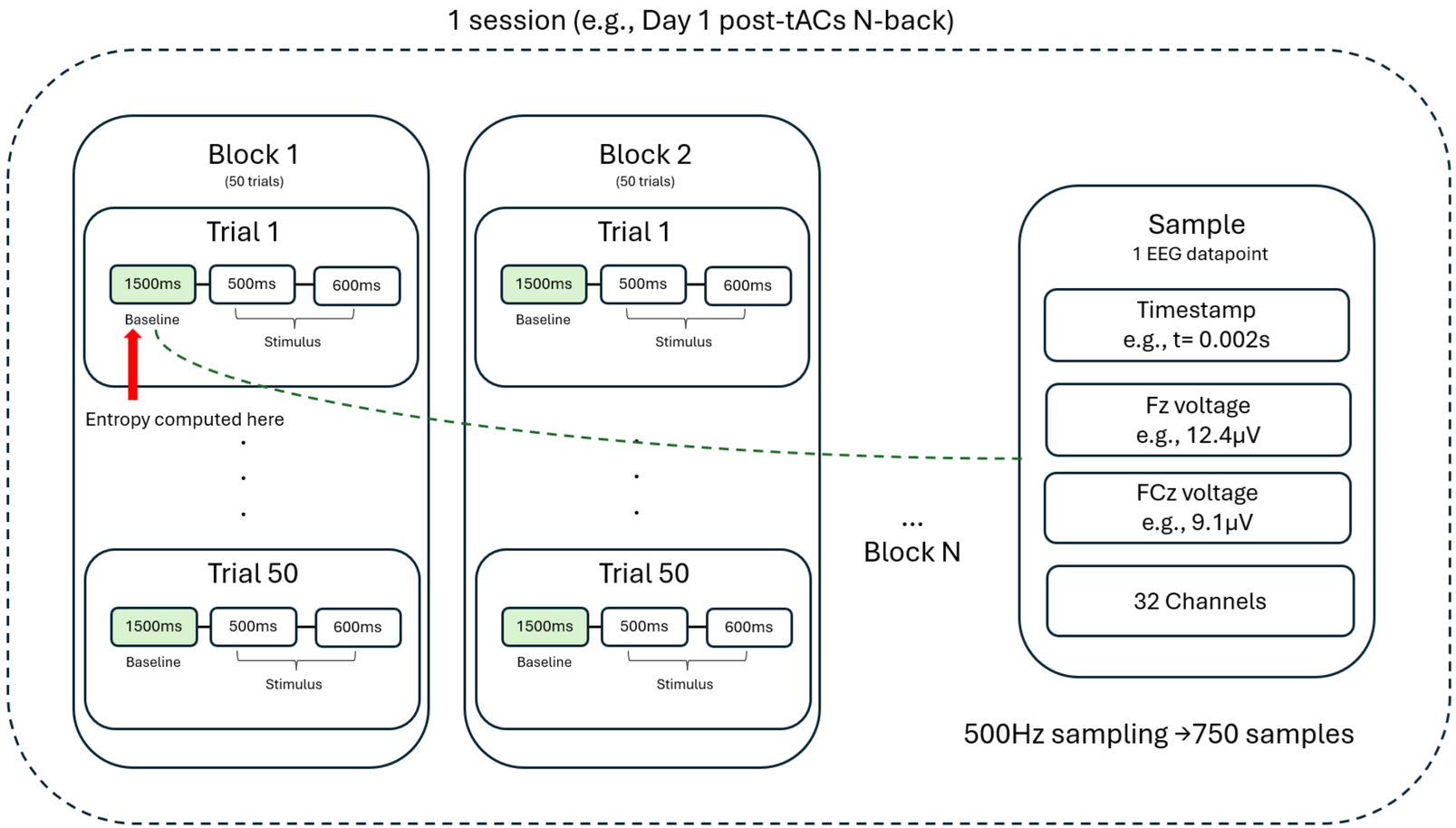

What we compare?

Pre-tACS N-back Baseline entropy

vs.

Post -tACS N-back Baseline entropy

Preprocessing and Analysis

Step One: Individualized Theta Frequency (ITF)

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Method: Modal Frequency Analysis

- Extract peak frequencies in 4-8 Hz range across time windows and frontal electrodes (Fz, FCZ) for Day 1 data

Step Two: Standardized Preprocessing Pipeline (Day 1 + Day 2)

Method: Standardized Preprocessing Pipeline

- Filter 1-50 Hz band-pass & 60 Hz Notch remove noise/drift
- Re-referencing for neutral baseline
- Artifact Removal (eye blinks & muscle activity)

Step Three: Analysis of BCI Learning

Method: Mixed-Effect ANOVA

- Feature Extraction (PSD) using Welch's method for 4-8Hz at Fz/FCz each BCI run
- Compare Active v. Sham groups for the runs

(Hsieh & Ranganath, 2014; Yeh et al., 2022; Welch, 1967)

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Preprocessing and Analysis

Step Four: EEG Entropy Analysis & Neural Complexity

Method: Statistical Test (Post – Pre) [8, 9]

- Extract 1500 ms inter-trial intervals from Day 1
- Calculate Shannon and Sample entropy for pre v. post-tACS
- Compare entropy between groups with independent sample t-tests

Step Five: BCI Decoding (Recalled v. Not Recalled)

Method: Spectral Biomarker Classification [10]

- Biomarkers: with focus on spectral pattern of increased Theta (4-8 Hz), decreased Alpha (8 –20 Hz), and increased Gamma (40- 128 Hz)
- Train LDA classifier to find recalled v. not recalled trail from those features

(Tian et al., 2017; Lau et al., 2022)

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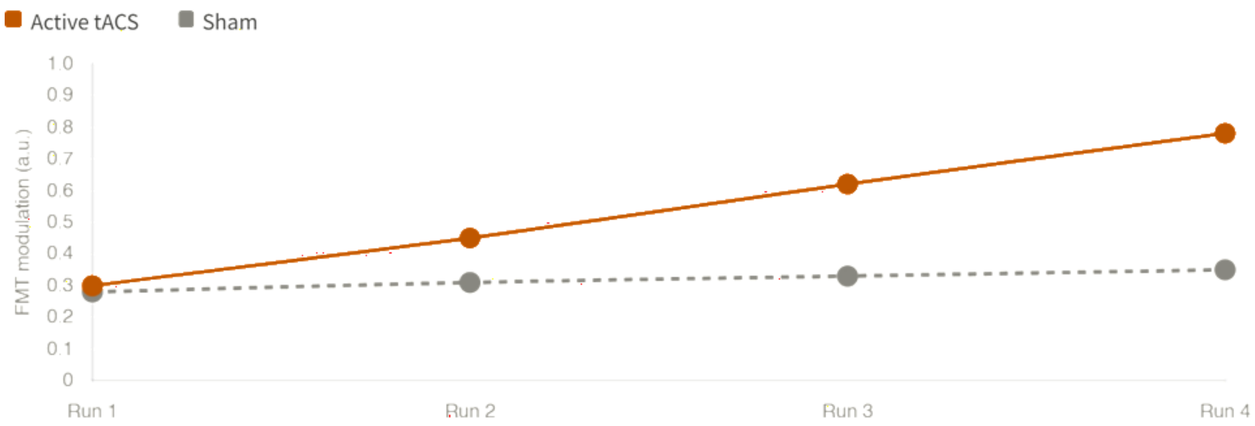
Predicted Outcomes

With our hypothesis, we would expect:

- Greater FMT modulation across Day 2 BCI runs in the active group
- Greater pre-to-post entropy change in inter-trial baseline windows in the active group
- Possibly modest improvements in N-back behavior, such as accuracy or reaction time

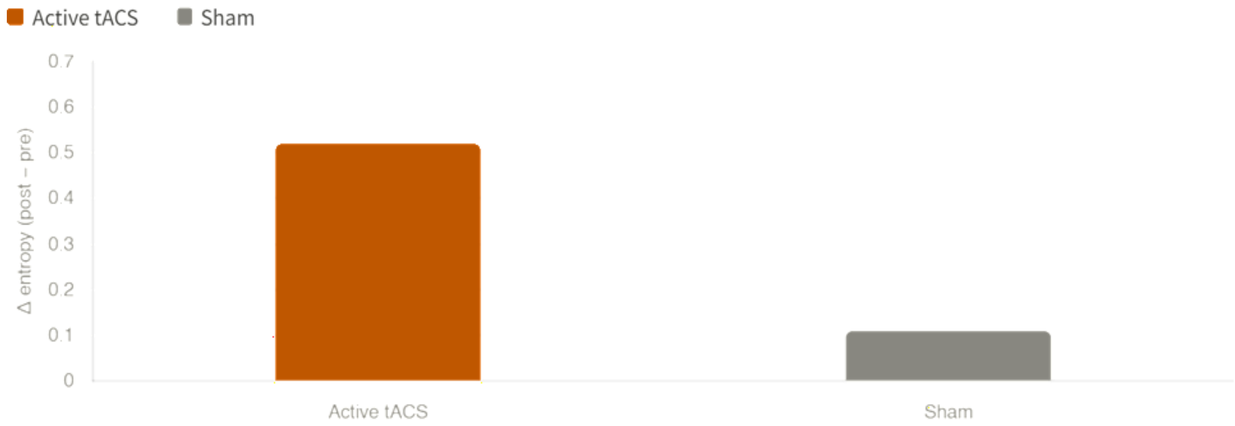
Prediction 1

FMT modulation across BCI runs (Day 2)



Prediction 2

Δ EEG entropy (post – pre tACS)



(Hsieh & Ranganath, 2014; Tian et al., 2017; Lau et al., 2022)

Predicted Outcomes and Exploratory Follow-Up

Does pre-to-post entropy change predict Day 2 BCI learning?

Δ Entropy
(Pre → Post tACS)

↓

Correlate / Regression

↓

Day 2 BCI
FMT Modulation

If supported:

Entropy may serve as a biomarker of tACS responsiveness

↓

Could help identify who benefits most from stimulation-enhanced BCI training

(Hsieh & Ranganath, 2014; Tian et al., 2017; Yeh et al., 2022)

Limitations

Small Sample Size	Limited participants and results are preliminary
Short Baseline Segments	Each inter-trail baseline is 1500 ms
Novel Approach	Using entropy to assess tACS effects on inter-trail baseline is new making it a risk

Resources

[1] Zeng, L., Guo, M., Wu, R., Luo, Y., & Wei, P. (2022). The Effects of Electroencephalogram Feature-Based Transcranial Alternating Current Stimulation on Working Memory and Electrophysiology. *Frontiers in aging neuroscience*, *14*, 828377. https://doi.org/10.3389/fnagi.2022.828377

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Resources

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[8] Janssen, T. W., Bink, M., Weeda, W. D., Geladé, K., van Mourik, R., Maras, A., & Oosterlaan, J. (2017). Learning curves of theta/beta neurofeedback in children with ADHD. European child & adolescent psychiatry, 26(5), 573-582
[9] Katerman, B. S., Li, Y., Pazdera, J. K., Keane, C., & Kahana, M. J. (2022). EEG biomarkers of free recall. NeuroImage, 246, 118748. https://doi.org/10.1016/j.neuroimage.2021.118748

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